

A. EXPERIMENT ENVIRONMENT

Hardware. Training was conducted on 1 NVIDIA 4090 GPU workstation with 48 GB of memory. The machine was equipped with an Intel(R) Xeon(R) Platinum 8470Q processor and 96GB of RAM, running Ubuntu22.04.

Software. Our implementation uses Python 3.12, PyTorch 2.8, and CUDA 12.8, with Transformers 5.14.1 and PyTorch Lightning 2.6.5. Detailed software requirements can be found in our repository.

B. LIST OF HYPER-PARAMETERS

Table 5 summarizes the settings for generator adaptation, artist-condition learning, and artist-classifier training. Layer indices are zero-based. Hyperparameters were chosen manually based on a small number of preliminary experiments. No systematic hyperparameter search was performed.

C. EVALUATION DETAILS

C.1. Metric Definitions and G.T. References

Artist Attribution Rate, Confidence, and Rank. We use an independently trained artist classifier with a frozen MERT backbone, learned layer-combination weights, and an MLP classification head. AAR is the fraction of generated clips whose prediction matches the source artist. Artist Conf. is the mean softmax probability assigned to that artist, and Artist Rank is its mean one-based rank among all classes, with ties handled by counting only strictly higher scores. For reference, we apply the same trained artist classifier to real recordings. The AAR reference is therefore ordinary top-1 accuracy, while Artist Conf. and Artist Rank describe the probability and rank of the true artist.

Ground-Truth Next-Token Confidence. GT-NTC measures the probability assigned to ground-truth EnCodec tokens during autoregressive generation. We use the actual sampling distribution after classifier-free guidance, temperature scaling, and any enabled sampling filters, while accounting for the codebook delay pattern. Unlike teacher-forced evaluation, subsequent predictions are conditioned on previously sampled tokens. Scores are averaged over valid codebooks and the evaluated rollout positions, excluding padding and the supplied audio prefix. Our implementation evaluates token positions 1, 5, 10, ..., 100.

MERT Similarity. We use the pretrained MERT-v1-95M² encoder, independently of the artist-classification head. Audio is converted to mono and resampled to 24 kHz. The final-layer representations are temporally averaged to obtain clip embeddings, and we compute cosine similarity between each generated clip and its corresponding reference. Higher scores indicate greater similarity in the pretrained music-representation space. For G.T. reference, we compare different songs by the same artist and songs by different artists sharing at least one coarse-genre label.

CLAP Score. We use the pretrained CLAP model³, with audio resampled to 48 kHz. CLAP score is the cosine similarity between

the projected audio embedding of a generated clip and the text embedding of its conditioning caption. For reference, the matched CLAP reference measures alignment between real audio and its own caption. The same-genre, different-artist reference pairs a song’s caption with audio from another artist in the same genre.

KL Divergence. We use a pretrained PaSST classifier to obtain softmax-normalized probabilities over 527 AudioSet event classes from reference and generated audio, resampled to 32 kHz. KL is computed in the reference-to-generated direction using natural logarithms and then averaged over paired clips. It measures differences in audio-event distributions, rather than artist-classification probabilities or MusicGen token predictions. Lower values indicate closer correspondence to the reference. The G.T. reference for KL Divergence is obtained exactly same as MERT similarity.

Fréchet Audio Distance. FAD compares distributions of 128-dimensional VGGish embeddings. Audio is converted to mono, normalized clip-wise, and processed at 16 kHz into approximately 0.96-second windows. We use pre-activation embeddings without PCA or quantization. For each artist, all windows from its generated clips and reference clips are collected into two sets, from which Gaussian means and covariances are estimated. We compute one FAD per artist and report their average. Thus, FAD is a set-to-set distance, not an average of point-to-point distances. For the FAD references, we sample size-matched real-audio sets from artist and genre scope, compute within-artist and cross-artist FAD respectively.

C.2. Additional Experimental Details

Inference Scenarios. All six scenarios use the same evaluation samples, source captions, and generation seeds. No Control uses the adapted generator without artist intervention. In continuation experiments, the supplied audio prefix is excluded from scoring, and generated continuations are compared with the corresponding reference segments reconstructed from cached EnCodec tokens.

Multi-artist suppression selects two to five distinct artists (randomly yet controlled by a fixed seed, making selection consistent across experiments). Target-inclusive sets contain the source artist, whereas non-target-only sets explicitly exclude it. Non-target selection refers to artist identity, not necessarily a different genre. Selected artist directions are individually decoded and peer-centered.

Negative-Only Learning. This ablation replaces Artist Direction Learning with a negative hinge objective that requires the controlled prediction loss to exceed the uncontrolled loss by a fixed margin. The objective becomes inactive once this margin is reached, which is exactly a mirrored version of our main objective.

Symmetry-Based Negative Supervision. Negative (Symmetry) instead matches the negative control’s increase in prediction loss to the improvement achieved by the matched positive control. The positive improvement is treated as a fixed and detached target. A `SmoothL1` loss supervises the difference between the negative effect and this target, with gradients passing only through the negative prediction. Unlike the negative hinge, symmetry penalizes both insufficient and excessive degradation relative to the positive improvement.

Other Ablations and Operating Points. The no-peer-residual ablation uses the raw decoded artist direction without subtracting the peer reference. Reg-off and Reg-on correspond to training without

²Model link: huggingface.co/m-a-p/MERT-v1-95M

³Model link: huggingface.co/laion/clap-htsat-unfused

Table 2. Ablation of the two key design choices. For every metric, the left and right sub-columns report single-target (1) and multiple-target (n) requests. All variants are trained without the specificity regularizer.

Variant	Style Similarity Measures										Generation Quality					
	AAR		Artist Conf.		Artist Rank		GT-NTC		MERT		CLAP $\uparrow$		KL $\downarrow$		FAD $\downarrow$	
	1	n	1	n	1	n	1	n	1	n	1	n	1	n	1	n
No control	0.422		0.362		8.583		0.039		0.879		0.492		0.318		1.517	
<i>Suppress target</i>																
Default (ours)	0.140	0.071	0.119	0.062	36.411	46.992	0.028	0.020	0.834	0.795	0.484	0.433	0.457	0.575	3.087	4.779
w/o peer residual	0.157	0.178	0.131	0.152	33.346	24.324	0.028	0.027	0.839	0.834	0.495	0.484	0.435	0.427	2.761	2.461
Negative-only	0.002	0.005	0.005	0.006	56.136	56.927	0.008	0.007	0.712	0.716	0.147	0.151	1.010	1.010	17.474	17.416
w/ negative (sym.)	0.162	0.082	0.135	0.072	33.760	43.826	0.029	0.021	0.841	0.800	0.484	0.445	0.426	0.538	2.764	4.220
<i>Suppress non-target</i>																
Default (ours)	0.321	0.169	0.273	0.143	13.634	24.109	0.033	0.023	0.854	0.811	0.478	0.434	0.361	0.503	1.988	3.730
w/o peer residual	0.319	0.263	0.275	0.223	12.940	15.055	0.033	0.029	0.856	0.840	0.484	0.481	0.366	0.391	1.968	2.224
Negative-only	0.005	0.004	0.006	0.006	58.290	57.943	0.005	0.007	0.709	0.715	0.146	0.152	1.007	1.011	17.416	17.183
w/ negative (sym.)	0.342	0.194	0.284	0.161	12.801	22.272	0.034	0.025	0.859	0.817	0.479	0.442	0.351	0.476	1.949	3.221

and with Specificity Regularization, respectively, rather than toggling the regularizer during inference. Their training weights and branch schedules are reported in Appendix B.

D. ABLATION STUDY

We examine two key design choices: (1) peer-reference subtraction (Sec. 3.1) and (2) artist direction learning without explicit negative supervision. Table 2 compares the unregularized default against removing peer-reference subtraction, negative-only training, and the gain-matched symmetry objective of Sec. 3.2.

Peer-reference subtraction. Removing peer-reference subtraction weakens target suppression, especially for multiple controls: AAR increases from 0.071 to 0.178, while artist rank decreases from 46.992 to 24.324. Notably, the separation between on- and off-target suppression narrows, particularly in the multiple-target scenario. This supports the role of peer subtraction for artist-specific control.

Explicit negative supervision. Negative-only training achieves near-zero target AAR, but non-target AAR also collapses to near-zero, and CLAP falls to extremely low values across the four scenarios. This pattern indicates non-specific degradation rather than selective artist suppression, illustrating why low attribution alone is insufficient evidence of successful control. Symmetry-based supervision yields a more conservative trade-off, with modest quality and off-target benefits at the expense of suppression strength. It does not demonstrate a clear overall advantage over direction-only learning, and we therefore retain the simpler direction-only objective.

E. STATISTICAL TEST OF RESULTS

Statistical tests were conducted separately for each evaluation metric at the artist level ($N = 64$). Clip-level measurements were first averaged within each artist to avoid treating clips from the same artist as independent observations. Because the same artists were evaluated under all six control scenarios, we used a one-factor repeated-measures ANOVA as the omnibus test. Greenhouse–Geisser-corrected degrees of freedom and p -values were used to account for potential violations of sphericity. A Friedman test was additionally conducted as a non-parametric robustness analysis. All eight metrics showed significant omnibus scenario effects under both tests (all $p_{GG} < 0.001$; all Friedman $p < 0.001$).

Following the omnibus tests, all $\binom{6}{2} = 15$ scenario pairs were compared using two-sided paired sign-flip permutation tests with 10,000 Monte Carlo permutations. The resulting p -values were adjusted separately within each metric using Holm’s procedure to control the family-wise error rate. A pairwise comparison was classified as significant only when both the Greenhouse–Geisser-corrected omnibus test and its Holm-adjusted pairwise test satisfied $\alpha = 0.05$. Fig. 4 and Fig. 5 presents the statistical comparisons across inference scenarios, with specificity regularization disabled and enabled. Please note that repeated values of $p_{Holm} = 0.0015$ in figures reflect the minimum attainable Holm-adjusted p -value under 10,000 Monte Carlo permutations and should not be interpreted as identical effect magnitudes.

F. DATASET ANALYSIS

Dataset composition. Table 3 summarizes the selected dataset, which contains 8,093 songs from 128 artists, corresponding to 38,570 audio clips and 311.76 hours of audio. Each artist contributes 40–64 songs (63.23 on average). The number of clips per artist varies more widely, from 64 to 984, with an average of 301.33. Partitions are song-disjoint rather than artist-disjoint: all clips from a song remain in the same partition, while all 128 artists are represented in each partition.

Table 3. Dataset size and song-disjoint partitions. Hours are summed from individual clip durations.

Split	Artists	Songs	Clips	Hours
Training	128	7,329	35,048	283.34
Validation	128	382	1,829	14.77
Test	128	382	1,693	13.64
Total	128	8,093	38,570	311.76

Genre coverage. Figure 2 shows the distribution of 15 coarse-genre labels and their composition within individual artists. Labels are non-exclusive: 4,776 songs (59.01%) carry more than one coarse-genre label. Ambient/new age and electronic/dance are the most

Table 4. Real-audio MERT cosine similarity across distinct-song pairs. Each row contains 50,000 sampled pairs. SD describes variability across pairs, not uncertainty in the mean.

Pair type	Mean	SD	Median	5th–95th percentile
Same artist, different songs	0.790	0.092	0.802	[0.619, 0.915]
Same genre, different artists	0.752	0.089	0.764	[0.584, 0.874]
Different genres and artists	0.745	0.083	0.757	[0.591, 0.860]

prevalent labels, covering 31.21% and 28.52% of songs, respectively, followed by cinematic (23.96%) and classical/orchestral (23.44%). The artist–genre heatmap also reveals overlapping genre profiles across artists, rather than a one-to-one correspondence between artist identity and coarse genre. Genre coverage is therefore reported at the song level, independently of the number of clips extracted from each song.

Artist relationships in MERT space. To examine acoustic relationships without over-weighting songs that yield more clips, we randomly select one clip per song using a fixed seed, obtaining 8,093 clips. We extract embeddings from the first 10 seconds of each selected clip using the frozen MERT-v1-95M encoder and compute cosine similarity. We consider three types of distinct-song pairs: the same artist; different artists sharing at least one coarse-genre label; and different artists with no shared coarse-genre label. For each type, we uniformly sample 50,000 unique unordered pairs without replacement and report micro-averaged statistics. The inter-artist matrix in Fig. 3 instead averages all cross-artist clip-pair similarities for each artist pair, with the diagonal excluded.

As shown in Table 4, songs from the same artist have higher average similarity (0.790) than songs from different artists, including those sharing a genre label (0.752). In contrast, the difference between same-genre and different-genre cross-artist pairs is small (0.752 versus 0.745), and their empirical distributions overlap substantially. These observations suggest that coarse-genre membership alone provides limited separation between artists in this representation space. The overlap with same-artist pairs further indicates that MERT similarity should not be interpreted as a direct measure of artist identity, supporting its use alongside the in-domain artist attribution metrics.

Table 5. Training configurations for generator adaptation, artist-condition learning, and artist-classifier training.

Parameter	Value
Shared Training Settings	
Optimizer	AdamW
Adam coefficients (β_1, β_2)	(0.9, 0.999)
Optimizer numerical stability ϵ	10^{-8}
Learning-rate schedule	Constant; no warm-up
Gradient clipping	Global norm, threshold 1.0
Gradient accumulation	None
Training precision	FP16 mixed precision
Artist-balanced sampling	Enabled
Generator Adaptation	
Pretrained generator	MusicGen-small
Trainable parameters	LoRA parameters only; backbone frozen
LoRA target projections	Cross-attention key and value projections
Adapted decoder layers	18–23
LoRA rank r / scaling coefficient α	32 / 32
LoRA dropout	0.05
LoRA initialization	Kaiming-uniform A ; zero-initialized B
Learning rate / weight decay	6×10^{-5} / 0
Batch size / maximum epochs	16 / 2
Training objective	Token cross-entropy with delayed-codebook teacher forcing
Artist-Direction Learning	
Generator initialization	Adapted generator from the preceding stage
Frozen parameters	All generator parameters, including LoRA
Trainable parameters	Artist condition bank, depth embeddings, shared decoder, and per-layer gates
Artist / depth / controller dimensions	512 / 512 / 512
Shared MLP decoder widths	$1024 \rightarrow 512 \rightarrow 1024 \rightarrow 512 \rightarrow 1024$, with GELU between linear layers
Intervention layers	12–23
Initial gate value / gate magnitude bound	0 / 1.0
Peer reference	Similarity-weighted aggregation over all non-self artists
Peer softmax temperature	0.2
Learning rate / weight decay	8×10^{-5} / 10^{-5}
Batch size / maximum epochs	16 / 30
Direction-learning margin / loss weight	0.5 / 1.0
Specificity weight: Reg-off / Reg-on	0 / 1.0
Specificity tolerance or margin	0.03
Training-branch schedule: Reg-off / Reg-on	N/A / Once upon five steps
In-domain Artist-Classifer Training	
Pretrained representation model	MERT-v1-95M; backbone frozen
Trainable parameters	Layer-combination weights and classification head
Feature aggregation	Learned softmax-weighted layer combination, followed by masked temporal mean pooling
Classification head	Layer normalization and a one-hidden-layer MLP with GELU
Hidden dimension / dropout	1024 / 0.2
Audio input	Mono, 24 kHz, up to 10 s
Audio cropping	Random crop for training; center crop for evaluation
Learning rate / weight decay	3×10^{-4} / 0.01
Batch size / maximum epochs	8 / 5
Training objective	Artist-classification cross-entropy

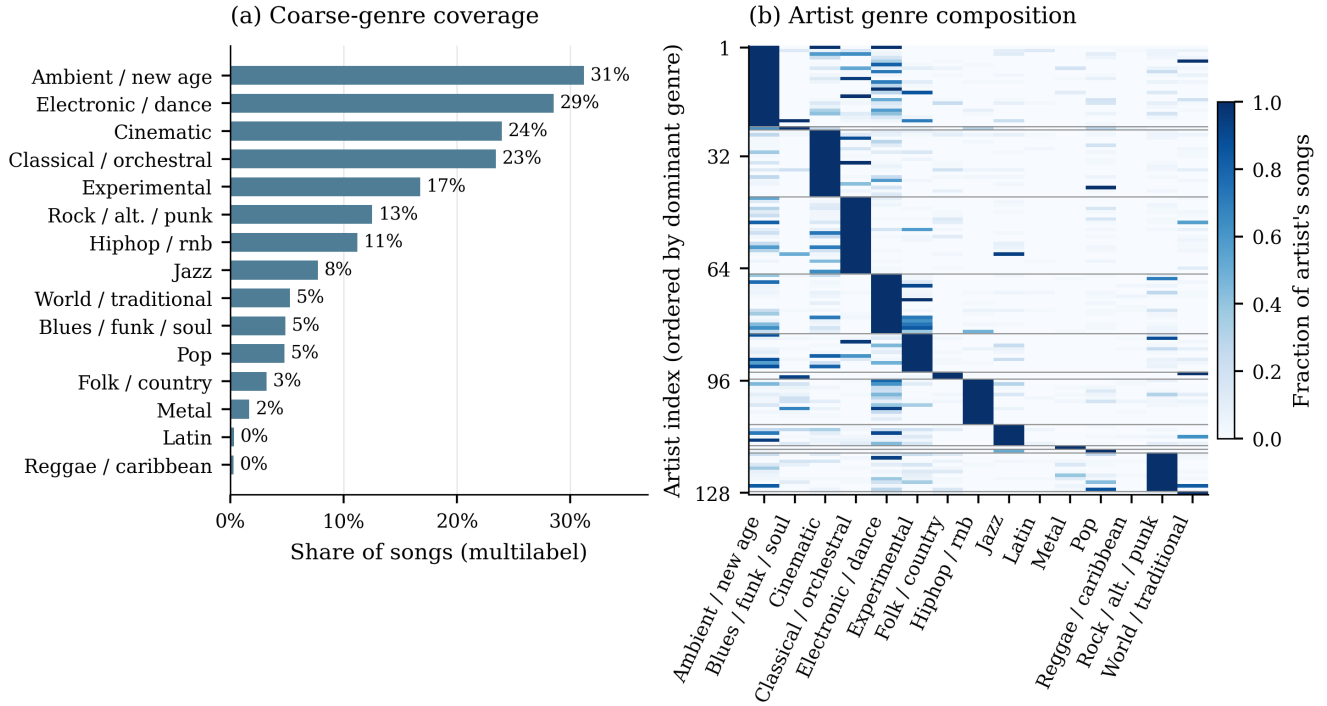


Fig. 2. Genre composition of the selected dataset. (a) Fraction of songs carrying each coarse-genre label. Labels are non-exclusive, so percentages need not sum to 100%. Displayed bar annotations are rounded to whole percentages. (b) Fraction of each artist's songs carrying each label. Artists are ordered by their most frequent genre label, with horizontal lines marking changes in that label.

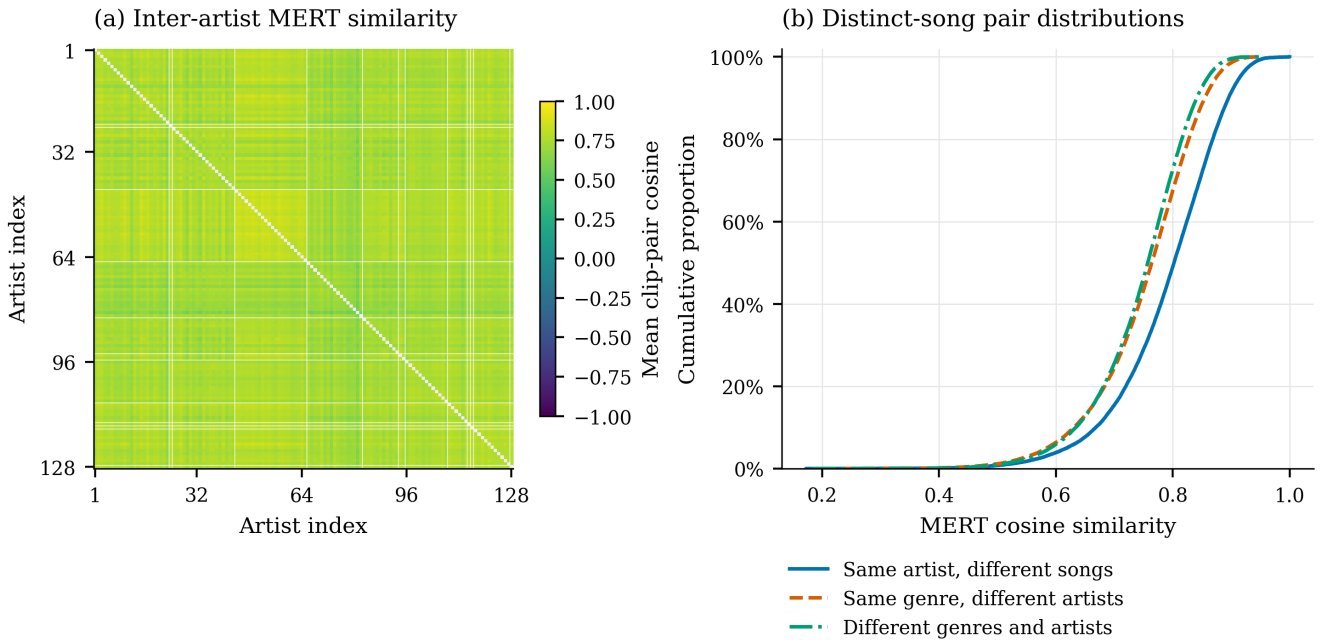
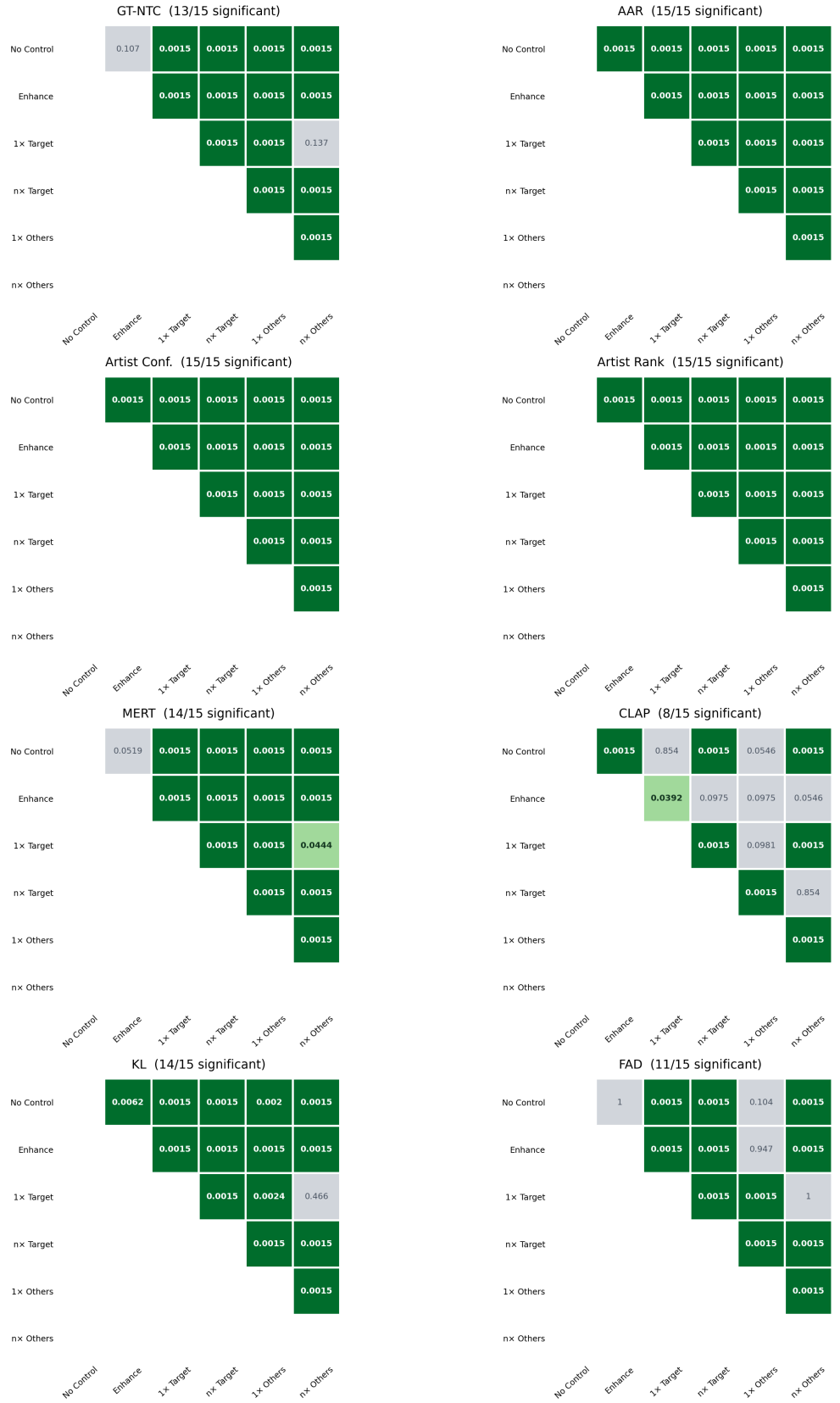


Fig. 3. Artist relationships measured on real audio. (a) Mean cross-artist clip-pair MERT cosine similarity, using the same artist ordering as Fig. 2. The diagonal is excluded; separators mark dominant-genre groups. (b) Empirical cumulative distributions for the three distinct-song pair types. A rightward shift indicates higher similarity. Both cross-artist distributions overlap strongly, whereas same-artist pairs tend to have higher similarity.



Fig. 4. Statistical comparisons across inference scenarios with specificity regularization disabled (Reg-off).



Not significant (omnibus + Holm, $\alpha=0.05$) $0.01 \leq \text{Holm-adjusted } p < 0.05$ Holm-adjusted $p < 0.05$

Fig. 5. Statistical comparisons across inference scenarios with specificity regularization enabled (Reg-on).